Class Project

Nutritional Analytics from Code-switched Gastronomy Influencer Reels Using Prompt Engineering and Large Language Models

DATABASE DESIGN DOCUMENT

BITP 3123 Distributed Application Development

Author

Emaliana Kasmuri

Fakulti Teknologi Maklumat dan Komunikasi

Universiti Teknikal Malaysia Melaka

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Description

This document describes the complete database schema for the nutritional LLM analysis system. The schema covers data collection, transcription, verification, ground truth annotation, experiment execution, and LLM output storage. Eleven tables are defined across three functional groups: data collection, ground truth, and experiment results.

The script to create the tables is available in ulearn.

Legend: PK = Primary Key FK = Foreign Key

1.1 Table Relationships

All foreign key relationships between the 11 tables.

From Table	To Table	Cardinality	Description
influencer	reel	1 to many	Each influencer is associated with multiple reels collected from their Instagram profile.
reel	audio_file	1 to 1	Each reel produces exactly one extracted audio file during the download pipeline.
reel	transcript	1 to 1	Each reel produces exactly one transcript generated by the Faster-Whisper transcription pipeline.
audio_file	transcript	1 to 1	Each audio file serves as the direct source for one transcript.
transcript	ground_truth_reel	1 to 1	Each transcript has exactly one corresponding reel-level ground truth annotation.
ground_truth_reel	ground_truth_ingredient	1 to many	Each annotated reel contains one or more

			ingredient-level ground truth entries.
transcript	experiment	1 to many	Each transcript is processed across multiple experimental conditions (up to 16 in Phase 1).
llm_model	experiment	1 to many	Each LLM model participates in multiple experiment runs across different transcripts and techniques.
prompt_technique	experiment	1 to many	Each prompt technique is applied across multiple experiment runs.
experiment	nutrition_result	1 to 1	Each experiment run produces exactly one structured nutrition result.
nutrition_result	ingredient_result	1 to many	Each nutrition result contains one or more individual ingredient-level extraction records.

Table: influencer

Stores the profile of a single Malaysian gastronomy influencer selected by the group during the early phase of the project. This table contains exactly one record.

Relationships

a. influencer (1) → reel (many)

Field Name	Data Type	Key	Description
influencer_id	INT	PK	Auto-increment primary key
name	VARCHAR(100)		Full name of the influencer
instagram_account	VARCHAR(100)		Instagram handle e.g. khairulaming
instagram_url	VARCHAR(500)		Full URL to the Instagram profile
created_at	TIMESTAMP		Record creation timestamp

Table: reel

Stores each Instagram Reel link identified by all group members during data collection. Each row represents one unique reel sourced from the insta-links Excel sheet, with each group member required to identify exactly 10 reels. Depending on group configuration of 4 to 5 students, this table must contain between 40 and 50 records, no fewer and no more.

Relationships

a. reel (many) → influencer (1)
b. reel (1) → audio_file (1)
c. reel (1) → transcript (1)

Field Name	Data Type	Key	Description
reel_id	INT	PK	Auto-increment primary key
influencer_id	INT	FK	References influencer.influencer_id
reel_id_instagram	VARCHAR(50)		Instagram reel ID extracted from URL e.g. DV7uZzBE47j
reel_url	VARCHAR(500)		Full Instagram Reel URL

identified_by_matric	VARCHAR(20)		Matric number of team member who identified the reel
identified_by_name	VARCHAR(100)		Name of team member who identified the reel
identified_date	DATE		Date the reel was identified
created_at	TIMESTAMP		Record creation timestamp

Table: audio_file

Stores file metadata and human verification results for each audio file extracted from a reel. Each reel produces exactly one audio file, so the total number of records in this table must correspond exactly to the number of records in the **reel** table.

Relationships

a. audio_file (many) → reel (1)
b. audio_file (1) → transcript (1)

Field Name	Data Type	Key	Description
audio_id	INT	PK	Auto-increment primary key
reel_id	INT	FK	References reel.reel_id
file_name	VARCHAR(200)		Audio file name e.g. DV7uZzBE47j.mp3
file_path	VARCHAR(500)		Full path to the audio file on disk
file_created_at	TIMESTAMP		File creation timestamp from filesystem metadata
file_size_bytes	BIGINT		Audio file size in bytes
file_format	VARCHAR(10)		Audio format e.g. mp3
reel_audio_consistent	BOOLEAN		True if audio matches the Instagram Reel content
verified_by_matric	VARCHAR(20)		Matric number of the audio verifier
verified_by_name	VARCHAR(100)		Name of the audio verifier

verified_at	TIMESTAMP		Timestamp when audio verification was completed

Table: transcript

Stores verified transcript text produced by Faster-Whisper, together with its execution metadata and human verification results. Each audio file and its associated reel produce exactly one transcript, so the total number of records in this table must correspond exactly to the number of records in both the **audio_file** and **reel** tables.

Relationships

a. transcript (many) → reel (1
b. transcript (1) → audio_file (1)
c. transcript (1) → ground_truth_reel (1)
d. transcript (1) → experiment (many)

Field Name	Data Type	Key	Description
transcript_id	INT	PK	Auto-increment primary key
reel_id	INT	FK	References reel.reel_id
audio_id	INT	FK	References audio_file.audio_id
file_name	VARCHAR(200)		Transcript file name e.g. transcription_20260406_081309.txt
file_path	VARCHAR(500)		Full path to the transcript file on disk
file_created_at	TIMESTAMP		File creation timestamp from filesystem metadata
file_size_bytes	BIGINT		Transcript file size in bytes
file_format	VARCHAR(10)		File format e.g. txt
audio_transcript_consistent	BOOLEAN		True if transcript matches the audio content
verified_by_matric	VARCHAR(20)		Matric number of the transcript verifier
verified_by_name	VARCHAR(100)		Name of the transcript verifier

verified_at	TIMESTAMP		Timestamp when transcript verification was completed

Table: ground_truth_reel

Stores reel-level ground truth annotation as produced by human annotators. Each record corresponds to one transcript and mirrors the Reels tab in the annotation Excel sheet. The total number of records must correspond exactly to the number of records in the transcript table, as every transcript requires one ground truth annotation.

Relationships

a. ground_truth_reel (many) → transcript (1)
b. ground_truth_reel (1) → ground_truth_ingredient (many)

Field Name	Data Type	Key	Description
gt_reel_id	INT	PK	Auto-increment primary key
transcript_id	INT	FK	References transcript.transcript_id
annotator_matric	VARCHAR(20)		Matric number of the annotator
annotator_name	VARCHAR(100)		Name of the annotator
annotated_at	TIMESTAMP		Timestamp when annotation was completed

Table: ground_truth_ingredient

Stores ingredient-level ground truth annotation for each reel. Each record corresponds to one annotated ingredient and mirrors the Ingredients tab in the annotation Excel sheet. One reel may contain multiple ingredients, so this table will have more records than **ground_truth_reel**.

Relationships

ground_truth_ingredient (many) → ground_truth_reel (1)

Field Name	Data Type	Key	Description
gt_ingredient_id	INT	PK	Auto-increment surrogate primary key.
gt_reel_id	INT	FK	References ground_truth_reel.gt_reel_id; links

			this ingredient to its parent reel annotation.
name_original	VARCHAR(200)		Ingredient name exactly as spoken in the transcript, referred to as **Identified Name** in the annotation sheet.
language_mentioned	VARCHAR(5)		Language used to mention the ingredient name in the transcript: Malay (MY), English (EN), or Others (OT).
name_en	VARCHAR(200)		English equivalent of the ingredient name, referred to as **English Name** in the annotation sheet.
quantity_expression	VARCHAR(200)		Faithful transcription of how the quantity was expressed in the video, preserving the speaker's original wording including code-switching, regional variants, and informal expressions. No translation or normalisation is applied (e.g., dua sudu besar, sikit je).
quantity_category	VARCHAR(50)		Category that best describes the quantity expression, selected from the controlled vocabulary in Appendix A. Determines how the expression is later converted to grams or millilitres.
quantity_unit_culinary	VARCHAR(100)		Standardised culinary unit recorded using the controlled vocabulary in Appendix B. Populated only when quantity_category is metric, culinary_measured, or culinary_unmeasured. Null for vague categories (vague_small, vague_medium, vague_large, taste_based, not_mentioned).

quantity_value_culinary	FLOAT		Actual count or measurement number extracted directly from the transcript. Decimals are permitted (e.g., 0.5, 1.5, 2.5).
estimated_weight_g	FLOAT		Ingredient weight in grams.
calories	FLOAT		Ground truth caloric value for this ingredient in kilocalories (kcal).
total_fat_g	FLOAT		Ground truth total fat content in grams.
saturated_fat_g	FLOAT		Ground truth saturated fat content in grams.
cholesterol_mg	FLOAT		Ground truth cholesterol content in milligrams.
sodium_mg	FLOAT		Ground truth sodium content in milligrams.
total_carbohydrate_g	FLOAT		Ground truth total carbohydrate content in grams.
dietary_fiber_g	FLOAT		Ground truth dietary fiber content in grams.
total_sugars_g	FLOAT		Ground truth total sugars content in grams.
protein_g	FLOAT		Ground truth protein content in grams.
vitamin_d_mcg	FLOAT		Ground truth vitamin D content in micrograms.
calcium_mg	FLOAT		Ground truth calcium content in milligrams.
iron_mg	FLOAT		Ground truth iron content in milligrams.
potassium_mg	FLOAT		Ground truth potassium content in milligrams.
annotation_layer	VARCHAR(10)		Annotation layer indicator: layer1 = faithful transcription; layer2 = standardised unit conversion.

annotator_matric	VARCHAR(20)		Matric number of the annotator who completed this record.
annotator_name	VARCHAR(100)		Full name of the annotator who completed this record.
annotated_at	TIMESTAMP		Timestamp recording when the annotation was submitted.

Table: llm_model

Reference table storing the four LLMs. Records are inserted once during system setup and remain static throughout the experiment. This table contains exactly four records, one for each model: Llama 3.1 8B, Mistral 7B, Qwen2.5 7B, and OpenBioLLM-8B.

Relationships

a. llm_model (1) → experiment (many)

Field Name	Data Type	Key	Description
model_id	INT	PK	Auto-increment primary key
model_name	VARCHAR(100)		Display name e.g. Llama 3.1 8B Instruct
model_tag	VARCHAR(100)		Ollama model tag e.g. llama3.1:8b
provider	VARCHAR(100)		Model provider e.g. Meta, Mistral AI, Alibaba, Saama AI
description	TEXT		Brief description of the model and its relevance to the study
created_at	TIMESTAMP		Record creation timestamp

Table: prompt_technique

Reference table storing the four prompt engineering techniques evaluated in the study. Records are inserted once during system setup and remain static throughout the experiment. This table contains exactly four records, one for each technique: zero-shot, few-shot, chain-of-thought, and structured-output. Prompt content is maintained in external text files; this table stores only the relative file paths.

Relationships

a. prompt_technique (1) → experiment (many)

Field Name	Data Type	Key	Description
technique_id	INT	PK	Auto-increment primary key
technique_name	VARCHAR(50)		Technique name e.g. zero-shot, few-shot, chain-of-thought, structured-output
system_prompt_file	VARCHAR(500)		Relative path to the system prompt file e.g. prompts/zero_shot_system.txt
user_prompt_file	VARCHAR(500)		Relative path to the user prompt template file e.g. prompts/zero_shot_user.txt
prompt_version	VARCHAR(10)		Version of the prompt files e.g. 1.0
description	TEXT		Brief description of the technique and its characteristics
created_at	TIMESTAMP		Record creation timestamp

Table: experiment

Central linking table representing one experimental run. Each row captures a unique combination of one transcript, one LLM model, and one prompt technique. With 4 models and 4 techniques, each transcript produces 16 experimental conditions in Phase 1, resulting in a total number of records equal to the number of transcripts multiplied by 16.

Relationships

a. experiment (many) → transcript (1)
b. experiment (many) → llm_model (1)
c. experiment (many) → prompt_technique (1)
d. experiment (1) → nutrition_result (1)

Field Name	Data Type	Key	Description
experiment_id	INT	PK	Auto-increment primary key
transcript_id	INT	FK	References transcript.transcript_id
model_id	INT	FK	References llm_model.model_id

technique_id	INT	FK	References prompt_technique.technique_id
rag_enabled	BOOLEAN		False for Phase 1 (prompt engineering only), True for Phase 2 (RAG)
status	VARCHAR(20)		Execution status: pending, running, completed, failed
executed_at	TIMESTAMP		Timestamp when the experiment was executed

Table: nutrition_result

Stores the structured nutritional output produced by the LLM for each experiment run. Each experiment produces exactly one nutrition result, so the total number of records in this table must correspond exactly to the number of records in the experiment table. Nutrition values are recorded at both per-serving and whole-recipe levels, matching the nutrition facts sheet format. The raw LLM response is preserved for debugging and auditability.

Relationships

a. nutrition_result (1) → experiment (1)
b. nutrition_result (1) → ingredient_result (many)

Field Name	Data Type	Key	Description
result_id	INT	PK	Auto-increment primary key
experiment_id	INT	FK	References experiment.experiment_id
recipe_name	VARCHAR(200)		Recipe name extracted by the LLM
servings_estimated	INT		Number of servings estimated by the LLM
serving_calories	FLOAT		Calories per serving (amount_per_serving)
serving_total_fat_g	FLOAT		Total fat per serving in grams
serving_saturated_fat_g	FLOAT		Saturated fat per serving in grams
serving_cholesterol_mg	FLOAT		Cholesterol per serving in milligrams

serving_sodium_mg	FLOAT		Sodium per serving in milligrams
serving_carbohydrate_g	FLOAT		Total carbohydrate per serving in grams
serving_fiber_g	FLOAT		Dietary fiber per serving in grams
serving_sugars_g	FLOAT		Total sugars per serving in grams
serving_protein_g	FLOAT		Protein per serving in grams
serving_vitamin_d_mcg	FLOAT		Vitamin D per serving in micrograms
serving_calcium_mg	FLOAT		Calcium per serving in milligrams
serving_iron_mg	FLOAT		Iron per serving in milligrams
serving_potassium_mg	FLOAT		Potassium per serving in milligrams
total_calories	FLOAT		Total calories for the full recipe
total_fat_g	FLOAT		Total fat for the full recipe in grams
total_saturated_fat_g	FLOAT		Total saturated fat for the full recipe in grams
total_cholesterol_mg	FLOAT		Total cholesterol for the full recipe in milligrams
total_sodium_mg	FLOAT		Total sodium for the full recipe in milligrams
total_carbohydrate_g	FLOAT		Total carbohydrate for the full recipe in grams
total_fiber_g	FLOAT		Total dietary fiber for the full recipe in grams
total_sugars_g	FLOAT		Total sugars for the full recipe in grams
total_protein_g	FLOAT		Total protein for the full recipe in grams
total_vitamin_d_mcg	FLOAT		Total vitamin D for the full recipe in micrograms

total_calcium_mg	FLOAT		Total calcium for the full recipe in milligrams
total_iron_mg	FLOAT		Total iron for the full recipe in milligrams
total_potassium_mg	FLOAT		Total potassium for the full recipe in milligrams
raw_json_output	TEXT		Original raw JSON response from the LLM for debugging
json_valid	BOOLEAN		True if the LLM output was valid parseable JSON
created_at	TIMESTAMP		Record creation timestamp

Table: ingredient_result

Stores each individual ingredient extracted by the LLM as part of a nutrition result. One nutrition result may contain multiple ingredient records, so this table will have more records than **nutrition_result**.

Relationship

a. ingredient_result (many) → nutrition_result (1)

Field Name	Data Type	Key	Description
ingredient_id	INT	PK	Auto-increment primary key
result_id	INT	FK	References nutrition_result.result_id
name_original	VARCHAR(200)		Ingredient name as spoken in transcript
name_en	VARCHAR(200)		English translation of the ingredient name
quantity_value	FLOAT		Numeric quantity as extracted by LLM
unit_original	VARCHAR(100)		Unit as extracted by LLM e.g. sudu besar
unit_en	VARCHAR(100)		English translation of the unit

estimated_weight_g	FLOAT		Estimated weight in grams as computed by LLM
calories	FLOAT		Estimated calories (kcal)
total_fat_g	FLOAT		Estimated total fat in grams
saturated_fat_g	FLOAT		Estimated saturated fat in grams
cholesterol_mg	FLOAT		Estimated cholesterol in milligrams
sodium_mg	FLOAT		Estimated sodium in milligrams
total_carbohydrate_g	FLOAT		Estimated total carbohydrate in grams
dietary_fiber_g	FLOAT		Estimated dietary fiber in grams
total_sugars_g	FLOAT		Estimated total sugars in grams
protein_g	FLOAT		Estimated protein in grams
vitamin_d_mcg	FLOAT		Estimated vitamin D in micrograms
calcium_mg	FLOAT		Estimated calcium in milligrams
iron_mg	FLOAT		Estimated iron in milligrams
potassium_mg	FLOAT		Estimated potassium in milligrams

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